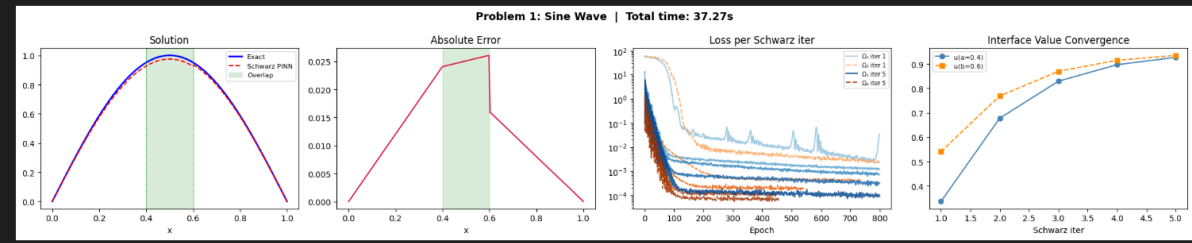
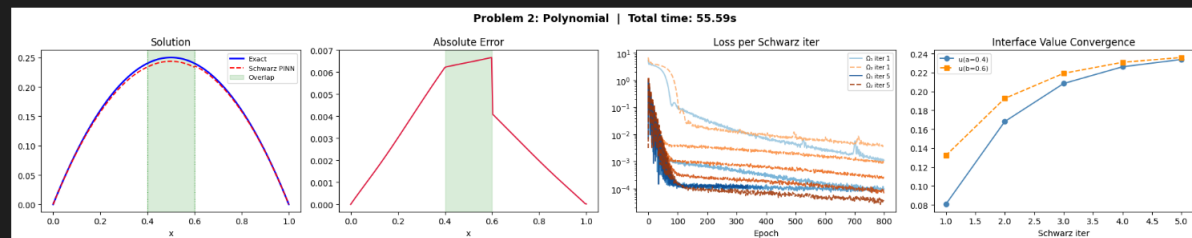


DDANN 1D (schwarz using soft constraints)

```
— Schwarz iter 1/5 —  
PINN1 (800 epochs, 5.71s) | u(a=0.4) = 0.336536 | loss = 0.034599  
PINN2 (800 epochs, 5.47s) | u(b=0.6) = 0.540882 | loss = 0.002072  
  
— Schwarz iter 2/5 —  
PINN1 (800 epochs, 5.34s) | u(a=0.4) = 0.677530 | loss = 0.001183  
PINN2 (735 epochs, 3.50s) | u(b=0.6) = 0.768593 | loss = 0.000453  
  
— Schwarz iter 3/5 —  
PINN1 (800 epochs, 3.33s) | u(a=0.4) = 0.829328 | loss = 0.000723  
PINN2 (543 epochs, 2.51s) | u(b=0.6) = 0.869908 | loss = 0.000183  
  
— Schwarz iter 4/5 —  
PINN1 (800 epochs, 3.55s) | u(a=0.4) = 0.896910 | loss = 0.000327  
PINN2 (455 epochs, 2.17s) | u(b=0.6) = 0.914971 | loss = 0.000109  
  
— Schwarz iter 5/5 —  
PINN1 (800 epochs, 3.72s) | u(a=0.4) = 0.926993 | loss = 0.000088  
PINN2 (456 epochs, 1.97s) | u(b=0.6) = 0.935017 | loss = 0.000061  
  
Total training time: 37.27s  
Max Absolute Error : 0.026055  
Mean Absolute Error: 0.012971  
Relative L2 Error : 0.021786
```



```
— Schwarz iter 1/5 —  
PINN1 (800 epochs, 4.46s) | u(a=0.4) = 0.080432 | loss = 0.001118  
PINN2 (800 epochs, 7.20s) | u(b=0.6) = 0.132031 | loss = 0.004349  
  
— Schwarz iter 2/5 —  
PINN1 (800 epochs, 7.19s) | u(a=0.4) = 0.168048 | loss = 0.000098  
PINN2 (800 epochs, 5.58s) | u(b=0.6) = 0.192560 | loss = 0.000883  
  
— Schwarz iter 3/5 —  
PINN1 (770 epochs, 6.50s) | u(a=0.4) = 0.208316 | loss = 0.000095  
PINN2 (800 epochs, 6.65s) | u(b=0.6) = 0.219107 | loss = 0.000268  
  
— Schwarz iter 4/5 —  
PINN1 (390 epochs, 3.19s) | u(a=0.4) = 0.226019 | loss = 0.000113  
PINN2 (800 epochs, 6.09s) | u(b=0.6) = 0.230783 | loss = 0.000067  
  
— Schwarz iter 5/5 —  
PINN1 (337 epochs, 2.97s) | u(a=0.4) = 0.233780 | loss = 0.000130  
PINN2 (800 epochs, 5.76s) | u(b=0.6) = 0.235880 | loss = 0.000037  
  
Total training time: 55.59s  
Max Absolute Error : 0.006667  
Mean Absolute Error: 0.003301  
Relative L2 Error : 0.021616
```



```

— Schwarz iter 1/5 —
PINN1 (800 epochs, 6.85s) | u(a=0.4) = 0.980698 | loss = 0.018848
PINN2 (487 epochs, 4.75s) | u(b=0.6) = -0.324074 | loss = 0.018348

— Schwarz iter 2/5 —
PINN1 (800 epochs, 7.10s) | u(a=0.4) = 0.760734 | loss = 0.010346
PINN2 (800 epochs, 5.92s) | u(b=0.6) = -0.470807 | loss = 0.003810

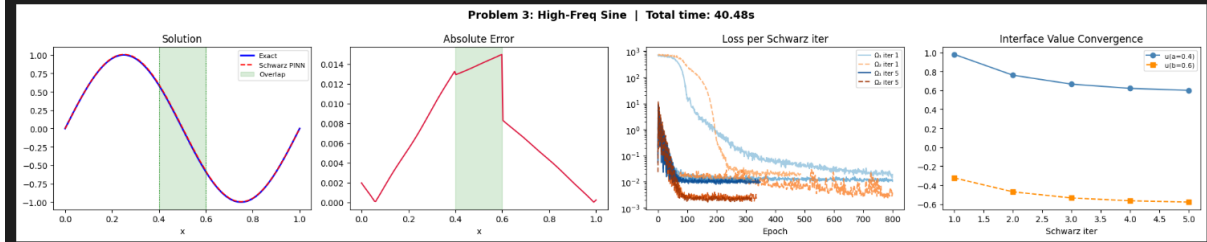
— Schwarz iter 3/5 —
PINN1 (321 epochs, 2.43s) | u(a=0.4) = 0.666381 | loss = 0.011813
PINN2 (318 epochs, 2.64s) | u(b=0.6) = -0.535566 | loss = 0.002201

— Schwarz iter 4/5 —
PINN1 (287 epochs, 2.29s) | u(a=0.4) = 0.621487 | loss = 0.011632
PINN2 (336 epochs, 2.45s) | u(b=0.6) = -0.565241 | loss = 0.002194

— Schwarz iter 5/5 —
PINN1 (349 epochs, 3.08s) | u(a=0.4) = 0.601161 | loss = 0.009507
PINN2 (321 epochs, 2.97s) | u(b=0.6) = -0.579443 | loss = 0.002276

Total training time: 40.48s
Max Absolute Error : 0.014983
Mean Absolute Error: 0.006777
Relative L2 Error : 0.011732

```



```

— Schwarz iter 1/5 —
PINN1 (800 epochs, 6.18s) | u(a=0.4) = 0.275469 | loss = 0.034091
PINN2 (800 epochs, 7.83s) | u(b=0.6) = 1.011161 | loss = 0.013110

— Schwarz iter 2/5 —
PINN1 (800 epochs, 7.20s) | u(a=0.4) = 0.950405 | loss = 0.001877
PINN2 (800 epochs, 8.59s) | u(b=0.6) = 1.460982 | loss = 0.000282

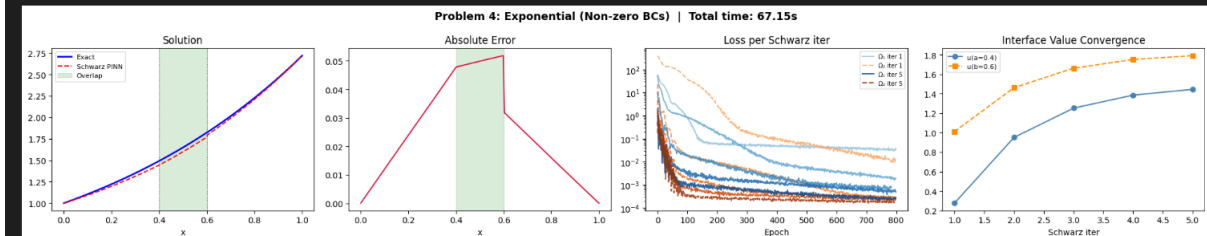
— Schwarz iter 3/5 —
PINN1 (785 epochs, 6.91s) | u(a=0.4) = 1.251000 | loss = 0.000798
PINN2 (800 epochs, 6.28s) | u(b=0.6) = 1.661342 | loss = 0.000279

— Schwarz iter 4/5 —
PINN1 (800 epochs, 5.67s) | u(a=0.4) = 1.384682 | loss = 0.000488
PINN2 (800 epochs, 6.36s) | u(b=0.6) = 1.750478 | loss = 0.000225

— Schwarz iter 5/5 —
PINN1 (800 epochs, 5.83s) | u(a=0.4) = 1.444018 | loss = 0.000277
PINN2 (800 epochs, 6.30s) | u(b=0.6) = 1.790086 | loss = 0.000171

Total training time: 67.15s
Max Absolute Error : 0.051832
Mean Absolute Error: 0.025808
Relative L2 Error : 0.017091

```



```

--- Schwarz Iteration 1/5 ---
Interface convergence -> u(a): 0.2852, u(b): -0.1066
--- Schwarz Iteration 2/5 ---
Interface convergence -> u(a): 0.1694, u(b): 0.0231
--- Schwarz Iteration 3/5 ---
Interface convergence -> u(a): 0.2899, u(b): 0.0968
--- Schwarz Iteration 4/5 ---
Interface convergence -> u(a): 0.3502, u(b): 0.1362
--- Schwarz Iteration 5/5 ---
Interface convergence -> u(a): 0.3864, u(b): 0.1599
Max Absolute Error : 0.030896
Mean Absolute Error: 0.014290
Relative L2 Error : 0.021974

```

Problem 5: Complex Non-Zero BCs ($[-3,0]$) | Total time: 58.94s

